

SIDDHESH AMIT BHURKE

Artificial Intelligence & Machine Learning Engineer

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AI/ML Engineer with hands-on experience building and deploying production-grade Generative AI pipelines using large language models (LLMs), LangChain, Gemini, and FastAPI. Skilled in CNNs, NLP, MLOps, RESTful APIs, and end-to-end machine learning model development. Seeking roles in Generative AI Engineering, Machine Learning Engineering, or LLM application development.

PROFESSIONAL EXPERIENCE

AI/ML Engineer Intern | Neural Network Labs

Jun 2025 – Nov 2025

- Architected multi-agent GenAI pipelines for ad creative generation using Gemini 1.5 Flash/Pro, cutting copywriting turnaround by ~60% through automated critique and refinement loops.
- Designed layout intelligence system with Gemini Vision applying rule-of-thirds principles, improving ad placement accuracy by ~30%.
- Launched modular image generation workflow using Vertex AI Imagen with 3-tier fallback logic (Replicate SDXL, Flux, Hugging Face), reducing pipeline failure rate by 40%.
- Delivered survey generator API (FLAN-T5 + Gemini Flash) producing 4 question formats with JSON schema enforcement, eliminating 50% of manual QA effort.
- Scaled RESTful API infrastructure via FastAPI to 5,000+ requests/month with async execution, API key auth, and production-grade CORS.
- Standardized LangChain workflows across 6+ LLM applications; containerized all services via Docker for MLOps-ready deployment on DigitalOcean.
- Shipped the full VisualCraft AI suite (Ad, Meme, Infographic Generators) end-to-end — from prototyping to production — within a 6-month internship scope.

PROJECTS

ATTEND AI – Smart Attendance Management System | Streamlit · OpenCV · face_recognition · Resemblyzer · Supabase [GitHub Repo](#)

- Built an AI-powered attendance platform automating classroom attendance via face recognition and optional voice biometrics, eliminating manual roll calls and attendance fraud.
- Engineered multi-face detection and recognition pipeline (OpenCV, dlib, face_recognition) to generate and match facial embeddings for automatic, photo-based attendance logging, paired with Resemblyzer voice-embedding verification for secondary identity confirmation.
- Designed QR-code / join-code subject enrollment flow with a Supabase-backed relational schema (teachers, students, subjects, attendance logs) supporting role-based teacher/student access.

Generative Ad Creator – VisualCraft AI(Internship Deliverable) | Gemini 1.5 · Vertex AI Imagen · LangChain · FastAPI · Pydantic

- Architected 4-stage multi-agent pipeline (copy → critique → refine → layout) achieving ~95% uptime across 3 image generation fallbacks (Replicate SDXL, Hugging Face, FLUX).
- Applied Gemini Vision for rule-of-thirds structured metadata extraction; deployed as async FastAPI microservice reducing layout generation errors.

GenAI Survey Question Generator – VisualCraft AI(Internship Deliverable) | FLAN-T5 · Gemini Flash · FastAPI · Prompt Engineering · Pydantic

- Designed NLP pipeline generating 10 structured questions per request across 4 formats; retry-safe logic cut API failure rate by ~40%.
- Deployed via FastAPI with JSON schema enforcement and correlation ID tagging, supporting reliable integration across 3+ client applications.

Glassdoor Salary Prediction System | XGBoost · Scikit-learn · Streamlit · Pandas · Joblib [GitHub Repo](#)

- Trained XGBoost regression on 50,000+ job listings, achieving ~18% accuracy improvement after hyperparameter tuning and feature engineering.
- Shipped interactive Streamlit dashboard for real-time predictions; serialized full pipeline with Joblib for one-command deployment.

Credit Score Classification – Paisabazaar | XGBoost · Scikit-learn · Pandas · Matplotlib [GitHub Repo](#)

- Achieved ~92% classification accuracy on 3 credit score bands using engineered features across income, credit utilization, and repayment behavior.
- Improved model reliability via cross-validation and feature importance analysis, reducing edge-case misclassification by 15%.

EDUCATION

B.E. Artificial Intelligence and Data Science

Dr. D.Y. Patil College of Engineering, Pune | Savitribai Phule Pune University |

Aug 2023 – 2027 (Expected)

GPA: 9.3/10

H.S.C PCM + CS

P.A.E.S & Jr. College, Satara | MHT-CET: 96.7%ile | HSC Boards : 72.6%

June 2021-2023

TECHNICAL SKILLS

Programming: Python, C, C++, SQL, HTML, CSS, JavaScript

Machine Learning & AI: Supervised Learning, Decision Trees, Random Forest, XGBoost, KNN, K-Means, Neural Networks, CNNs, DNNs, NLP, Time Series Forecasting, Model Evaluation (ROC-AUC, F1-Score, Confusion Matrix)

Generative AI & LLMs: LLMs, Prompt Engineering, LangChain, Gemini 1.5 Flash/Pro/Vision, Vertex AI Imagen, Hugging Face Transformers, FLAN-T5, Replicate API, SAM, Multi-modal Text/Image Generation

Frameworks & Libraries: Scikit-learn, TensorFlow, Keras, FastAPI, Pandas, NumPy, Matplotlib, Seaborn, Pydantic, AsyncIO, OpenCV

Deployment & MLOps: DigitalOcean, Docker, Streamlit, Git, RESTful APIs, Jupyter Notebook, Google Colab, Swagger, Postman, Power BI, Cloudflare R2

ADDITIONAL INFORMATION

Certifications: LinkedIn Learning – Machine Learning and AI; IBM – The AI Ladder: A Framework for Deploying AI; ML Bootcamp; ML Workshop IIT Bombay; WebDev 45-Day Workshop

Languages: English (Fluent), Hindi (Fluent), Marathi (Native) | Hackathons: Participated in Generative AI hackathons and ML challenges

Leadership: Documentation Head, Vihang Kalamandal (Art & Theatre Club); national-level theatre competitions (Purushottam, Firodia, RajyaNatya, Symbiosis Karandak); Trinity College of Music, London Accredited